MIL-L-52825A(ME)

14 April 1983
SUPERSEDING
MIL-L-52825(ME)
9 July 1974

MILITARY SPECIFICATION

LADDER: HANGING

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a prefabricated aluminum hanging ladder.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5440

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-A-200/8	- Aluminum Alloy 6061, Bar, Rod, Shapes, Tube and Wire, Extruded.
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-621	- Box, Wood, Nailed and Lock-Corner.
PPP-B-636	- Box, Shipping, Fiberboard.
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

MILITARY

MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377	- Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-889	- Dissimilar Metals.

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MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MS35206	- Screw, Machine - Pan Head, Cross-Recessed, Carbon Steel, Cadmium Plated, Unc-2a.
MS35425	- Nut, Plain, Wing Unc-2b.

(Copies of specifications and standards required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The ladder shall be as shown in figure 1 and as specified herein.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.3).

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3.2.1 Material deterioration and control. The ladder shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.2.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.2.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.2.2 Aluminum tube. Aluminum tube shall conform to QQ-A-200/8, temper 6.

3.2.3 Aluminum plate. Aluminum plate shall conform to QQ-A-250/11, temper 6.

3.3 Machine screws. Machine screws shall conform to MS35206-287.

3.4 Wing nuts. Wing nuts shall conform to MS35425-39.

3.5 Strength. The ladder shall support a weight of not less than 800 pounds equally distributed on any rung while hanging from the hooks provided thereon.

3.6 Identification marking. Each ladder shall be identified in accordance with MIL-STD-130.

3.7 Workmanship.

3.7.1 Welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas (argon or helium) shielded-arc method. Welds may be ground, filed, buffed, or chipped but not hammered and shall be free from defects such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, porosity, lack of penetration, lack of prescribed fit, slag inclusion, or spatter.

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3.7.2 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS D1.1, Structural Welding Code, Section 5, Qualification.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ASME or AWS codes.

3.7.3 Weld quality.

a. The contractor shall be responsible for determining the inspections needed to insure that, when the welded parts are assembled together to make the ladder, the ladder shall conform to the inspection requirements specified in Section 4.
b. All weldments shall be free of slag, flux, weld spatter, and other impurities detrimental to both the appearance and strength of the weldment.
c. Undercut in weldments shall be held to a minimum but shall not be more than 1/32 inch deep.

3.7.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. When wing nuts are correctly torqued, threads shall not protrude more than two full threads.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves

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the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured and examined in accordance with referenced specifications and standards.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.3).
b. Inspection of packaging (see 4.5.)

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and test shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. AQL shall 0.4 percent defective for major defects and 4.0 percent defective for minor defects.

4.3.3 Test. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 1.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The ladders shall be examined as specified herein, for the following defects:

Major

101. Any part of the ladder not in accordance with figure 1.
102. Components missing.
103. Material not as specified.
104. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environment.
105. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
106. Contractor does not have documentation available for identification of material, material finishes or treatments.
107. Workmanship not as specified.

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Minor

201. Identification marking missing, illegible, or not as specified.

4.4.2 Test. The ladder shall be suspended from a support by the hooks provided. Each rung shall be subjected to a weight of not less than 800 pounds euqally distributed along its length. Breaking of any part of the ladder, weld failure, or permanent deformation of the hooks or any part of the ladder shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For purpose of inspection, a complete pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

108. Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
109. Disassembly not as specified for level A.
110. Components not bundled and secured together as specified for level A.
111. Quantities packed togehter exceed the weight limitation of the box for level A or B.
112. Preservation and packing not in accordance with the referenced document as specified for commercial.
113. Marking missing, illegible, incorrect or incomplete for level A, B, or commercial.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The ladder shall be disassembled to allow the most practicable minimum cubage by removal of the machine screws and wing nuts. The removed machine screws and wing nuts shall be placed together in a cotton, draw

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string, or plastic, heat sealed, bag. The components comprising the complete ladder shall then be secured together to form a compact bundle with tape conforming to PPP-T-97, type II, class A or B; and the contents placed in a close-fitting box conforming to PPP-B-636, W5c, style FOL. Box closure shall be in accordance with the appendix to the box specification.

5.1.2 Commercial. The ladder shall be preserved in accordance with MIL-STD-1188.

5.2 Packing. Packing shall be level A, B, or commerical, as specified (see 6.2).

5.2.1 Level A. Ladders, preserved as specified in 5.1, shall be packed together in a close-fitting box conforming to PPP-B-601, overseas type, style optional; or PPP-B-621, class 2, style optional, in quantities not to exceed the weight limitation of the box specification. The boxes shall be closed and strapped in accordance with the appendix to the applicable box specification.

5.2.2 Level B. Packing shall be as specified in 5.2.1 for level A, except the boxes shall be domestic type or class as applicable.

5.2.3 Commercial. Ladders preserved as specified in 5.1 shall be packed in accordance with MIL-STD-1188.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129.

5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The hanging ladder is intended for use in the erection of military bolted steel storage tanks.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Degree of preservation and degree of packing required (see 5.1 and 5.2).

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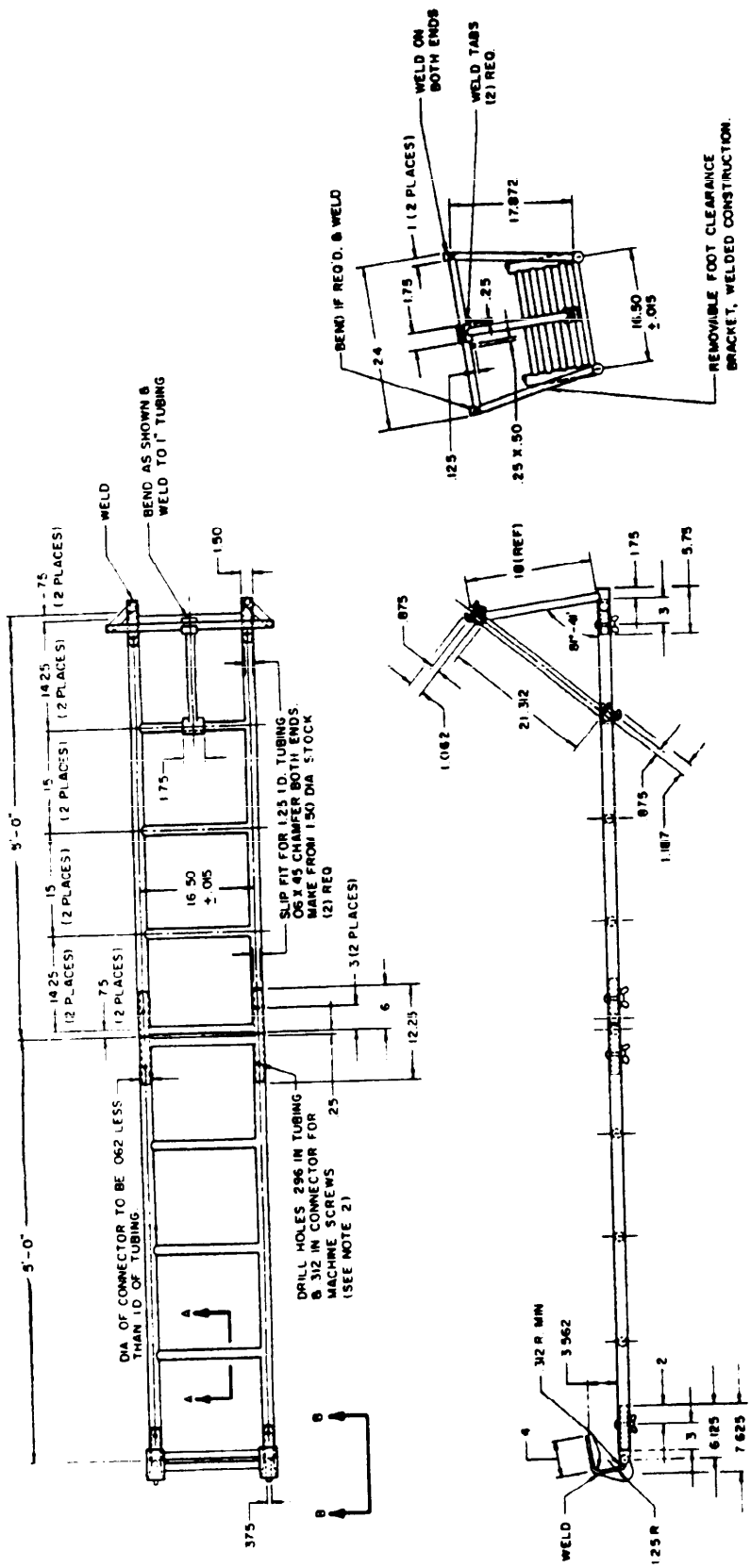
6.3 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.2).

Custodian:
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FIGURE 1. Ladder, hanging.

NOTES
1. ALL DIMENSIONS IN INCHES UNLESS OTHERWISE SPECIFIED.
2. HOLES FOR MACHINE SCREWS TO BE LOCATED WITHIN ±.015 OF NOMINAL DIMENSION & SHALL BE DIAMETRICALLY LOCATED PERPENDICULAR TO THE PLANE OF THE LADDER

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-52825A(ME)	Ladder: Hanging

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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